DAY 50 - EXPERT

Final Expert Capstone

Primary tool: Integrated Data Analysis

What you are learning

End-to-end expert system: problem framing, multi-source data, cleaning, SQL/Python/BI, QA, AI verification, governance and executive communication.

Learning objectives

- Build a complete analytical system

- Defend every KPI

- Automate repeatable steps

- Document lineage/governance

- Present management recommendations

Worked practical

Complete the final multi-source assessment and submit system, dashboard, QA report, AI-use log, executive brief and oral defense.

Source / Input	Method	Output / Check
Raw data	Apply the lesson skill	Result linked to source
Change one input	Recalculate / rerun	Confirm expected change
Control total / test	Validate	Explain the result

Guided practice

1. Recreate the lesson idea on a small example.
2. Change inputs and predict the result before recalculating.
3. Complete the supplied independent dataset in the required tool.
4. Create a control check or reconciliation.
5. Explain the result in ordinary language.

AI coaching

AI may assist with debugging, critique and testing, but every conclusion and calculation must be independently verified and explainable.

AI discipline

Attempt first. Use AI for explanation, hints, debugging, critique, test cases and mock questioning. Never submit output you cannot explain and independently verify.

Independent exercise

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Common failure modes

- Wrong grain or key.
- Formula/query/code answers a different question.
- Missing or unmatched records are silently ignored.
- Findings are not reconciled to the numbers.
- AI-generated logic is accepted without testing.

Mastery gate

[ ] I can reproduce the method without copying.
[ ] I can trace the result to the source.
[ ] I have a QA/control check.
[ ] I can explain a limitation.
[ ] I can answer an assessor's follow-up question without asking AI.